

RVHS 2016 Year 6 Prelim 2: 8864 H1 Mathematics Marking Scheme

Question 1 [3 marks]		
1	$2x^2 - m(6x - 5m) - 2$ $= 2x^2 - 6mx + 5m^2 - 2$ For $2x^2 - m(6x - 5m) - 2$ to be always positive, $(-6m)^2 - 4(2)(5m^2 - 2) < 0$ $36m^2 - 40m^2 + 16 < 0$ $-4m^2 + 16 < 0$ $m^2 - 4 > 0$ $(m - 2)(m + 2) > 0$ $m > 2 \text{ or } m < -2$	

Question 2 [5 Marks]**(a)**

$$y = \ln \left[(x+1)^2 (1-2x^2)^{-5} \right]$$

$$y = \ln (x+1)^2 + \ln (1-2x^2)^{-5}$$

$$y = 2 \ln (x+1) - 5 \ln (1-2x^2)$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{2}{x+1} - \frac{5(-4x)}{1-2x^2} \\ &= \frac{2}{x+1} + \frac{20x}{1-2x^2} \end{aligned}$$

(b)

$$\frac{1}{x+1} + \frac{1}{(x+1)^2} = \frac{x+1+1}{(x+1)^2} = \frac{x+2}{(x+1)^2}$$

$$\int_0^2 \frac{x+2}{(x+1)^2} dx = \int_0^2 \frac{1}{x+1} + \frac{1}{(x+1)^2} dx$$

$$= \left[\ln(x+1) - \frac{1}{x+1} \right]_0^2$$

$$= \ln 3 - \frac{1}{3} - (0 - 1)$$

$$= \ln 3 + \frac{2}{3}$$

Question 3 [6 Marks]**(i)**

$$y = a + e^{bx}$$

$$\frac{dy}{dx} = be^{bx}$$

$$\text{At } x = 0, \frac{dy}{dx} = b$$

Gradient of tangent = 3

Thus $b = 3$ Sub (0, 2) into $y = a + e^{3x}$

$$2 = a + 1$$

$$a = 1$$

(ii)

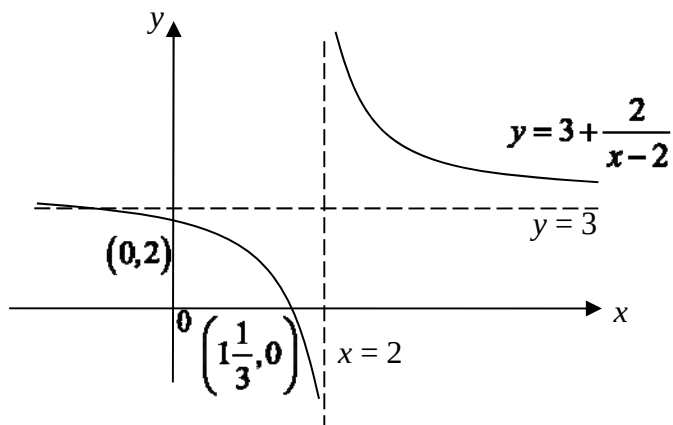
$$\text{Gradient of Normal} = -\frac{1}{3}$$

$$\text{Equation of Normal: } y - 2 = -\frac{1}{3}(x - 0)$$

$$y = -\frac{1}{3}x + 2$$

Question 4 [10 marks]

(i)



(ii)

$$y = 3 + \frac{2}{x-2}$$

$$y - 3 = \frac{2}{x-2}$$

$$x - 2 = \frac{2}{y-3}$$

$$x = 2 + \frac{2}{y-3}$$

$$\text{Area enclosed} = \int_5^8 \left(2 + \frac{2}{y-3} \right) dy$$

$$= \left[2y + 2 \ln |y-3| \right]_5^8$$

$$= [16 + 2 \ln 5] - [10 + 2 \ln 2]$$

$$= 6 + 2(\ln 5 - \ln 2)$$

$$= 6 + 2 \ln \frac{5}{2}$$

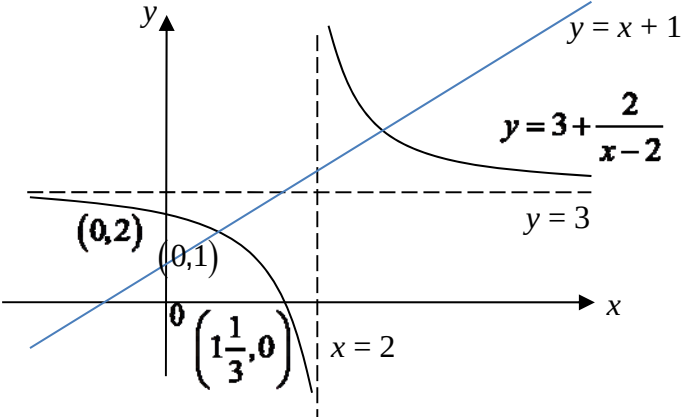
(iii)

$$\frac{2}{x-2} \leq x-2$$

$$\frac{2}{x-2} \leq x-3+1$$

$$3 + \frac{2}{x-2} \leq x+1$$

$$\text{Add } y = x + 1$$

	 $3 + \frac{2}{x-2} = x + 1$ $\frac{2}{x-2} = x - 2$ $2 = x^2 - 4x + 4$ $x^2 - 4x + 2 = 0$ $x = \frac{4 \pm \sqrt{16 - 4(1)(2)}}{2}$ $= \frac{4 \pm \sqrt{8}}{2}$ $= 2 \pm \sqrt{2}$ From the graph, $2 - \sqrt{2} \leq x < 2$ or $x \geq 2 + \sqrt{2}$	
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Question 5 [11 Marks]		
(a)	Let V be the volume and h the height of the fish tank. $V = x^2 h$ $2000 = x^2 h$ $h = \frac{2000}{x^2}$ Let A be the area of the four side walls and the base of the fish tank.	

$$A = x^2 + 4xh$$

Hence the cost of fish tank is:

$$\begin{aligned} C &= 2x^2 + (4xh) \\ &= 2x^2 + \left(4x \frac{2000}{x^2}\right) \\ &= 2x^2 + \frac{8000}{x} \end{aligned}$$

$$\frac{dC}{dx} = 4x - \frac{8000}{x^2}$$

When $\frac{dC}{dx} = 0$,

$$\frac{8000}{x^2} = 4x$$

$$x^3 = 2000$$

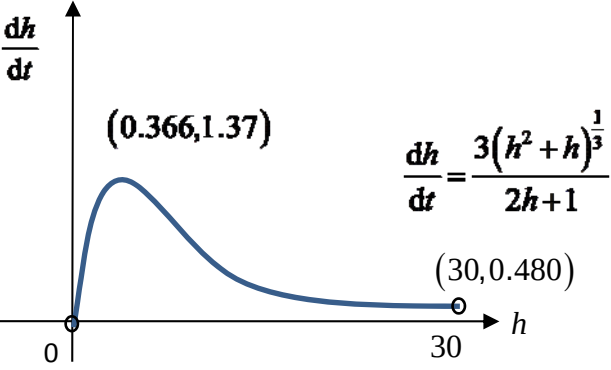
$$x = \sqrt[3]{2000}$$

$$\frac{d^2C}{dx^2} = 4 + \frac{16000}{x^3}$$

When $x = \sqrt[3]{2000}$, $\frac{d^2C}{dx^2} = 4 + \frac{16000}{2000} = 12 > 0$.

Alt

x	12.55	$\sqrt[3]{2000} \approx 12.6$	12.65
$\frac{dC}{dx}$	-0.5928	0	0.6070
			

(bi)	$\frac{dV}{dt} = 2 \text{ cm}^3 / \text{s}$ $V = (h^2 + h)^{\frac{2}{3}}$ $\frac{dV}{dh} = \frac{2}{3}(h^2 + h)^{-\frac{1}{3}}(2h + 1)$ $\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$ $2 = \frac{2}{3}(h^2 + h)^{-\frac{1}{3}}(2h + 1) \times \frac{dh}{dt}$ $\frac{dh}{dt} = \frac{3(h^2 + h)^{\frac{1}{3}}}{2h + 1}$ Substitute $h = 8$, $\frac{dh}{dt} = \frac{3(8^2 + 8)^{\frac{1}{3}}}{16 + 1} \approx 0.734$ $\therefore$ The rate of change of the depth of the water when $h = 8$ is 0.734 cm/s.	
(bii)	 The rate of change of depth of water increases from 0 to 1.37 cm/s then it decreases as the height increases.	

Question 6 [5 Marks]				
(i)	Quota Sampling. This method is non-random as not every teacher has an equal chance of being selected.			
(ii)	Stratified Sampling. Principal can draw random samples from each strata as follows:			
		Senior teachers	Beginning teachers	Classroom teachers
	Number of teachers selected	$0.05 \times 20 = 1$	$0.6 \times 20 = 12$	$20 - 1 - 12 = 7$

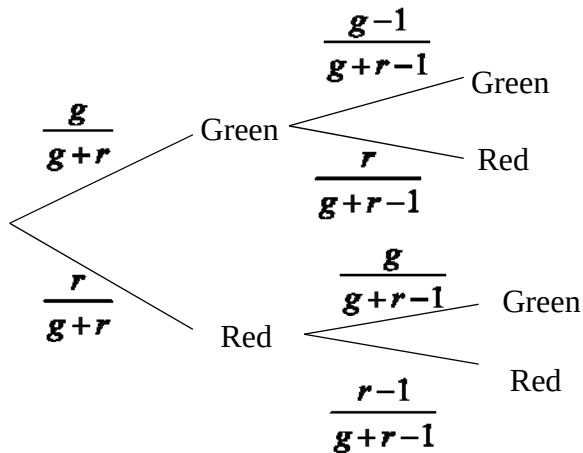
Question 7 [5 Marks]		
(i)	Since $n = 500$ is large, by CLT, $\bar{Y} \sim N\left(50724, \frac{13112^2}{500}\right)$ $P(\bar{Y} - 50724 \leq c) = 0.95$ $P(-c \leq \bar{Y} - 50724 \leq c) = 0.95$ $P(50724 - c \leq \bar{Y} \leq 50724 + c) = 0.95$ $P(\bar{Y} \leq 50724 - c) = 0.025$ $50724 - c \approx 49574.70364$ $c \approx 1150 \text{ (3 sig fig)}$	
(ii)	It is not necessary to assume population normal as $n = 500$ is large, thus sample mean is normally distributed by the CLT.	

Question 8 [8 Marks]		
(i)	Using the GC, $a \approx 10$	
(ii)	Let Y be the r.v. for the number of haircut that takes less than 10 minutes, $Y \sim B(n, 0.369441)$. $P(Y \leq 3) < 0.25$ Using the GC, $n = 12, P(Y \leq 3) = 0.296 > 0.25$ $n = 13, P(Y \leq 3) = 0.231 < 0.25$ $n = 14, P(Y \leq 3) = 0.179 < 0.25$ The least value of $n = 13$.	
(iii)	Let W be the r.v. of the number of haircut more than 10 minutes out of 100, $W \sim B(100, 1 - 0.369441)$. $W \sim B(100, 0.630559)$ Since $n = 100$ is large, $np = 100(0.630559) = 63.1 > 5$, $n(1-p) = 36.9 > 5$, thus $W \sim N(63.0559, 23.295)$ approximately. $P(W \geq 50) \stackrel{cc}{=} P(W \geq 49.5) = 0.998$	

Question 9 [10 marks]		
(i)	By symmetry, $\mu = \frac{45 + 65}{2} = 55$ $P(X < 45) = 0.1$ $P\left(Z < \frac{45 - 55}{\sigma}\right) = 0.1$ $\frac{-10}{\sigma} = -1.281551567$ $\sigma = 7.80304 \approx 7.80$	
(ii)	$Y \sim N(60, 11^2)$ $P(Y \geq a) \leq 0.1$ $P(Y < a) \geq 0.9$ $a \geq 74.1$ $\therefore$ Least raw score is 75	
(iii)	$Y - X \sim N(5, 11^2 + 7.80304^2)$ $P(Y \geq X + 10)$ $= P(Y - X \geq 10)$ $= 0.355$	
(iv)	The required probability $= 2P(X < 65)P(X > 65)$ $= 0.18$	

Question 10 [10 marks]

(a)(i)



(a)(ii)

P(2 balls of different colours)

$$= \left(\frac{g}{g+r} \right) \left(\frac{r}{g+r-1} \right) + \left(\frac{r}{g+r} \right) \left(\frac{g}{g+r-1} \right)$$

$$= \frac{2rg}{(r+g)(r+g-1)}$$

(b)(i)

Since $P(B|A) = \frac{1}{5} \neq P(B) = \frac{1}{4}$, A and B are not independent.

(b)(ii)

Given $P(A|B') = \frac{7}{10}$

$$\frac{P(A \cap B')}{P(B')} = \frac{7}{10}$$

$$P(A \cap B') = \frac{7}{10} P(B')$$

$$P(A \cap B') = \frac{7}{10} \left(1 - \frac{1}{4} \right) = \frac{21}{40}$$

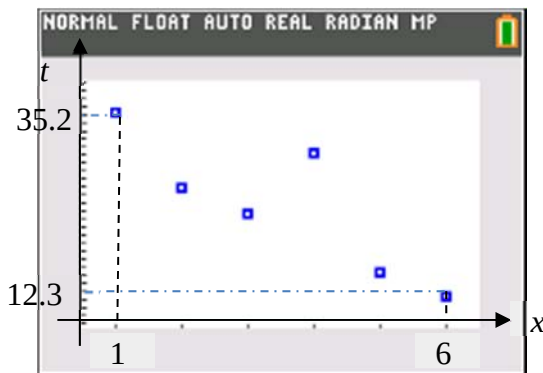
(b)(iii)

$$P(A \cup B) = P(A \cap B') + P(B)$$

$$P(A \cup B) = \frac{21}{40} + \frac{1}{4} = \frac{31}{40}$$

Question 11 [10 Marks]

(i)



(ii)

$$r \approx -0.851$$

Although $|r|$ is rather close to 1, suggesting a linear relationship but looking at the scatter diagram, you will notice that there is a outlier.

(iii)

$$(4, 30.1)$$

(iv)

$$r = -0.9750384 \approx -0.975$$

(v)

A suitable regression line:

$$t = 36.72209 - 4.24767x$$

$$\text{i.e. } t = 36.7 - 4.25x$$

When $x = 8$,

$$t = 36.72209 - 4.24767(8)$$

$$= 2.7406977 \approx 2.74 \text{ mins}$$

Although $|r|$ is close to 1, the estimate is not reliable because $x = 8$ is outside of data range, $[1, 6]$. Therefore it is a extrapolation.

(vi)

regression line t on x (i.e $t = ax + b$)should be use since x is the controlled or independent variable.

Question 12 [12 Marks]		
(i)	$\bar{x} = \frac{980}{100} = 9.8$ $s^2 = \frac{1}{99} \left[9700 - \frac{980^2}{100} \right] = \frac{32}{33}$	
(ii)	Let μ be the population mean of the mass of one packet of salt. $H_0 : \mu = 10$ $H_1 : \mu < 10$ Since $n = 100$ is large, Under H_0, $Z = \frac{\bar{x} - 10}{s/\sqrt{100}} \sim N(0,1)$ Using GC, the p-value = 0.0211 < 0.05 Reject the null hypothesis and there is sufficient evidence at 5% significance level to reject the company's claim that the mass is at least 10g.	
(iii)	For a 2-tailed test, the p-value = 2(0.0211) = 0.0422 < 0.05. Thus reject the null hypothesis and there is sufficient evidence at 5% significance level that the mass of the packets of salt it packs differs from 10 grams.	
(iv)	Let Y be the r.v. of the mass of each packet of salt in the new packaging system. $H_0 : \mu = 10$ $H_1 : \mu < 10$ Since $n = 60$ is large, Under H_0, $Z = \frac{\bar{Y} - 10}{\frac{0.9}{\sqrt{60}}} \sim N(0,1)$ Since company's claim is valid, H_0 is not rejected. $\frac{m - 10}{\frac{0.9}{\sqrt{60}}} \geq -2.3263$ $m \geq 9.7297 \approx 9.73$ Least possible value of m is 9.73.	

